INDIAN STATISTICAL INSTITUTE

Mid-Semester Examination (2025–26)

Programme: M.Math. 2nd Year

Course: Algebraic Geometry

Date: February 25, 2026

Time: 2 hours

Maximum Marks: 30

Instructions.

- This is **not** an open-notes examination; calculators are not allowed.
- Justify all steps in your arguments. Do not assume results not covered in lectures without providing a clear proof.
- Clarity and logical presentation are essential. Vague answers may lose marks. Partial solutions may receive some credit.

Notations & Conventions.

- Throughout, k denotes an algebraically closed field unless otherwise specified.

(1) Show that each of the following sets is not algebraic:
  (a) $\{(\cos t, \sin t, t) \in \mathbb{A}^3_{\mathbb{R}} : t \in \mathbb{R}\}$. [3 marks]
  (b) $\{(z, w) \in \mathbb{A}^2_{\mathbb{C}} : |z|^2 + |w|^2 = 1\}$, where $|x + iy|^2 = x^2 + y^2$ for $x, y \in \mathbb{R}$. [3 marks]

(2) (a) Let F be a nonconstant polynomial in $k[X_1, \ldots, X_n]$. Show that $\mathbb{A}^n_k \setminus \mathscr{Z}(F)$ is infinite for $n \geq 1$, and that $\mathscr{Z}(F)$ is infinite for $n \geq 2$. [4 marks]
  (b) Deduce that if $Z \subsetneq \mathbb{A}^n_k$ is a proper affine algebraic set, then $\mathbb{A}^n_k \setminus Z$ is infinite for all $n \geq 1$. [2 marks]

(3) Show that any conic in $\mathbb{P}^2_k$ is isomorphic to $\mathbb{P}^1_k$. [6 marks]

(4) Let C be an affine curve in $\mathbb{A}^3_k$. Prove that $\mathbb{A}^3_k \setminus C$ is never an affine variety. [6 marks]

(5) Let P_1, P_2, P_3 (respectively Q_1, Q_2, Q_3) be three non-collinear points in $\mathbb{P}^2_k$. Show that there exists a projective change of coordinates

$$T : \mathbb{P}^2_k \to \mathbb{P}^2_k$$

such that $T(P_i) = Q_i$ for $i = 1, 2, 3$. [6 marks]

(6) Prove that

$$\mathrm{Aut}_k(\mathbb{A}^1_k) = \{\, t \mapsto at + b \mid a \in k^\times,\ b \in k \,\}.$$

[6 marks]

(7) Give an explicit example of two affine curves C and D over k and a morphism

$$\varphi : C \to D$$

such that the induced map on Zariski topological spaces is a homeomorphism, but φ is not an isomorphism. [6 marks]

(8) Give an example of a morphism $\varphi : \mathbb{A}^2_k \to \mathbb{A}^2_k$ such that $\mathrm{Im}(\varphi)$ is dense but not open. Are all fibers equidimensional? Justify your answer. [6 marks]

* * * All the best * * *